

KUSHAL PANDYA

Ahmedabad, Gujarat

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PROFILE SUMMARY

Final-year Computer Engineering graduate (CGPA 8.97) with hands-on experience building end to end AI/ML systems across computer vision (YOLOv8), NLP, and LLM-powered applications (RAG, LangChain). Completed three internships covering full-stack development, classical ML, and production ML workflows. Seeking a fresher AI/ML Engineer role.

EXPERIENCE

Kody Technolab Ltd.

Jan 2026 – June 2026

Role – AI/ML Engineer Intern

Ahmedabad (On-site)

- Built, evaluated, and optimized Machine Learning (Regression, Classification, Clustering) and Deep Learning (ANN, CNN) models on real-world datasets; applied Hyperparameter Tuning (GridSearchCV, RandomizedSearchCV) to improve model F1-scores by up to 12%.
- Performed end-to-end EDA and data cleaning pipelines using Pandas, NumPy, Matplotlib, and Seaborn; delivered visualisation dashboards for stakeholder review.
- Optimised ML training throughput using Python multiprocessing and multithreading, reducing batch processing time by approximately 35%.

Infotact Solution

May 2025 – Jul 2025

Role – AI/ML Intern

Bengaluru (Remote)

- Built an AI task manager using NLP (Word2Vec embeddings) and ML classifiers (SVM, Random Forest) achieving 88% task-categorisation accuracy; system prioritised 500+ daily tasks.
- Developed a hybrid movie recommendation engine combining collaborative filtering and content-based methods on the MovieLens 100K dataset; improved RMSE from 1.02 (baseline) to 0.87.

SADHRASYA SoftTech

May 2024 – Jun 2024

Role – Web Developer

Vadodara, Gujarat

- Developed “Cravio,” a full-stack food ordering and real-time tracking app using the MERN stack.
- Integrated third-party APIs for payments, authentication, and live order tracking; reduced transaction failures by 30% through robust error handling and retry logic.

PROJECTS

RAG-Powered Document Q&A System

- Designed and deployed an end-to-end Retrieval-Augmented Generation (RAG) pipeline using LangChain, Hugging Face sentence-transformer embeddings, FAISS vector store, and Groq’s LLaMA3-8B model for contextual PDF Q&A.
- Enabled dynamic multi-PDF ingestion with semantic chunking; vector search retrieves top-k relevant passages with sub-second latency on a 50-page document corpus.
- Deployed as an interactive Streamlit web app; system handles natural language queries and returns cited, real-time LLM-generated answers demonstrating full model-to-UI deployment.

Unwanted Activity Detection Using YOLOv8

- Built a real-time anomaly detection system using YOLOv8 with transfer learning, fine-tuned on the Avenue Dataset (16 anomaly classes); achieved 95% frame-level detection accuracy vs. 67% SVM baseline.
- Built a custom data pipeline: automated frame extraction, bounding-box annotation, and augmentation (flip, HSV jitter) to generate a 12,000-frame training set from raw surveillance video.
- Optimised inference pipeline with OpenCV for real-time video processing at 28 FPS on CPU, enabling deployment without GPU infrastructure.

GujToGuj: AI-Powered Gujarati Speech-to-Text

- Built an automated speech recognition pipeline for Gujarati language audio/video, extracting and preprocessing audio from diverse formats (MP4, WAV) using MoviePy and SpeechRecognition.
- Fine-tuned a Transformer-based ASR model achieving 85% word-level accuracy on Gujarati speech a low-resource language with limited existing tooling.
- Reduced end-to-end transcription time by 40% vs. manual transcription baseline through batch audio segmentation and parallel processing.

TECHNICAL SKILLS

Languages: Python

ML / Deep Learning: Scikit-learn, TensorFlow, OpenCV, YOLO (v5/v8), CNN, RNN, Transformers, NLP, EDA, Regression, Classification, Clustering, Hyperparameter Tuning

GenAI / LLM: LangChain, FAISS, Hugging Face (Transformers, Sentence-Transformers), Groq API, LLaMA3, RAG pipelines, Streamlit

ML Libraries: Pandas, NumPy, Matplotlib, Seaborn

ML Mathematics: Statistics, Linear Algebra, Probability, Calculus

Developer Tools: Git, GitHub, VS Code, PyCharm, Jupyter Notebook, Microsoft PowerBI

EDUCATION

Charotar University of Science and Technology

B.Tech – Computer Engineering — CGPA: 8.97

Aug 2022 – Jun 2026

Anand, Gujarat

CERTIFICATIONS

- Microsoft Certified: Machine Learning Operations Engineer Associate(AI300)
- Microsoft Certified: SQL AI Developer Associate(DP800)
- Deep Learning – IIT Ropar (NPTEL) – **Elite Badge**
- IBM Data Science Professional Certificate (v3)

EXTRACURRICULAR

- **Smart India Hackathon (SIH) 2024** – Designed “Emission Niyantran,” an AI/ML web app (Streamlit + SQL) that predicts coal mine carbon emissions with **92% accuracy** and recommends reduction strategies.
- **DoseHack’24** – CHARUSAT with MediTab, Sep 2024: Developed a healthcare solution under a 24-hour hackathon constraint; collaborated with a cross-functional team on rapid prototyping.
- **Reimagine Hackathon** – Sheryians Coding School, Sep 2024: Built and deployed dynamic UI components with optimised backend logic, improving app responsiveness.